

DEWDRIVE · REQUIREMENTS

DewDrive requirements

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

Document ID	DWD-REQ-001	Version	0.3
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/dewdrive		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	TRL 3 update (design point and budget decided by Amish in DWD-DDR-001; R11 redefined to \$500; status column from DWD-CAL-001)
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with a design point and concept status
0.1	2026-09-24	Amish Chadha	Initial scaffold

DewDrive requirements

These requirements were checked by calculation at TRL 3 in DWD-CAL-001. The design as drawn meets eight of thirteen: R1 and R2 (water yield) are **not met**, R7 (salt containment) is at risk, and R6 and R10 can only be verified by test. Targets are still proposals for review, not user-validated needs, and will be revised after co-design sessions.

■ Design point

Decided by Amish on 2026-09-25 (DWD-DDR-001, D8). Unless a requirement says otherwise, it applies at this design point:

- Night: 10 h adsorption at 20 °C air and 40 % RH (6.92 g/m³ of vapour).
- Day: 6.0 kWh/m² of sun on the tilted aperture, of which 4.24 kWh/m² falls in the 09:00 to 15:00 window (DWD-CAL-001, C1; TRL 2 said 4.5); air at 35 °C.
- Low-humidity case: night air at 20 °C and 25 % RH, same sun.

■ Requirements

Table 1. Requirements and TRL 3 status. Values and tags are from DWD-CAL-001, Table 4.

ID	Requirement	Target	Verification	TRL 3 status
R1	Produce water at the design point	0.5 L per day or more collected, per unit	Cyclic mass and heat balance (DWD-CAL-001); later field test	Not met: 0.20 L per day as drawn (C3). About 0.53 L per day if night air is drawn through the bed (C5, proposed)
R2	Produce water in dry nights	0.25 L per day or more at 25 % night RH	Same	Not met: 0.09 L per day as drawn (C4). About 0.35 L per day with through-flow (C5, proposed)
R3	Use only the sun for regeneration	No fuel or grid power; bed reaches 75 °C or more by midday on a clear day at 35 °C ambient	Collector heat balance	Met: bed about 104 °C at noon (C2)
R4	Run its own electrical loads	Fan and logger need 30 Wh per day or less, supplied by the on-board PV panel, with 2 days or more of battery autonomy	Energy budget	Met: 23.6 Wh per day against 35 Wh per day; 2.6 days of autonomy (F1, F2)
R5	Keep the condenser cool	Condenser 15 K or less above ambient at the peak of desorption	Condenser heat balance	Met, thin margin: 13.1 K (C2)
R6	Keep the water clean	All wetted parts food-grade; no salt carry-over (chloride in the collected water below 250 mg/L)	Materials review; later laboratory water test	Not verifiable at TRL 3. Materials chosen to meet it
R7	Contain the salt	No liquid CaCl ₂ solution leaves the trays, even after a night at 90 % RH	Pore volume against solution volume; later chamber test	At risk: the pores fill at 51 % RH at equilibrium (E2); one 90 % RH night fills 46 % of them as drawn, 111 % with through-flow (E3)
R8	Be simple to operate	Two user actions per day (open flaps in the evening, close them in the morning), 5 min per day or less	Task analysis; co-design sessions	Met by design
R9	Be portable and quick to set up	Box 35 kg or less (two-person lift); stand separable; set up by two people in 30 min or less	Mass from model volumes; later trial	Met: box 30.9 kg, 24.0 kg with trays out; total 51.4 kg (G1, G2). Set-up time needs a trial
R10	Survive the site	Stable in 20 m/s (72 km/h) wind with anchors or ballast; UV-stable glazing; sorbent 300 cycles or more with 20 % or less loss of capacity	Wind load calculation; supplier data; later cycling test	Not verifiable at TRL 3. Wind: 49 kg of ballast or two anchors of 120 N each are needed (H2 to H4)
R11	Cost	Parts \$500 or less (<code>project.yaml</code> budget, raised from \$400 by Amish on 2026-09-25, DWD-DDR-001 D1)	Priced BOM	Met: \$488 (J1)
R12	Record performance	Log air temperature and RH, bed and condenser temperature every 5 min for 30 days or more; daily water mass recorded by weighing	Storage estimate; design review	Met: 8,640 records, 0.41 MB (I1)
R13	Protect users	External surfaces a person would touch in normal use stay at 60 °C or less; the	Surface temperature estimate; design review	Met: glazing outer skin about 53 °C at stagnation, fins up to 48 °C (D2, D3); 12.8 V DC

ID	Requirement	Target	Verification	TRL 3 status
		glazing is labelled hot; no exposed conductor above 60 V DC		

■ Assumptions

- The composite (about 33 wt % CaCl_2 in mesoporous silica gel) is modelled from bulk salt thermodynamics: at equilibrium it holds 0.507 g/g at 40 % RH and 0.330 g/g at 25 % RH (DWD-CAL-001, A4). The yield is set by how fast the bed takes up and gives off water, not by these capacities. Measured isotherms and uptake rates of the actual composite must replace the model.
- The day ends with the salt dried to about the dihydrate; the night starts with the bed at air temperature.
- One unit is a supplement: even about 0.5 L per day is about one fifth of one person's survival drinking need of 2.5 to 3 L per day (WHO technical note 9).

Safety: Requirements R6, R7, R10 and R13 exist because of hazards: untreated water, a corrosive salt solution, wind loads on a 1.1 m² panel, and hot surfaces. They cannot be waived without a documented decision by Amish.